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Topic 7 — Rates of Reaction and Energy Changes

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 2

Some reactions are over in a fraction of a second (an explosion, a precipitation), others take weeks (rusting) or years (weathering of limestone). This topic is about **how fast** reactions go and **why**, using **collision theory**, and about the **energy changes** that go with every chemical reaction — why some reactions warm their surroundings and others cool them, and how we can use **bond energies** and **reaction profiles** to explain this.

Content marked **(HT)** is Higher Tier only. Topic 7 is common to GCSE Chemistry and Combined Science — there is no chemistry-only content in this topic. It contains **one core practical**: investigating the effect of changing conditions on the rate of a reaction.

7.1 What Is the Rate of a Reaction?

> The **rate of a reaction** is how quickly reactants are used up or products are formed. It is measured as the **change in the amount (mass, volume or concentration) of a reactant or product per unit time**.

mean rate of reaction = amount of reactant used ÷ time taken

mean rate of reaction = amount of product formed ÷ time taken

Quantity measured	Unit of quantity	Unit of rate

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Volume of gas produced	cm^3	**cm^3/s** (or cm^3/min)
Mass lost (gas escapes) or mass of product	g	**g/s** (or g/min)
Concentration of a reactant/product	mol/dm^3	**$mol/dm^3/s$**
Time for a fixed change to happen (e.g. cross disappears)	s	rate ∝ **1/time** (unit s^{-1})

Worked Example: *(Illustrative data.)* Magnesium ribbon reacts with dilute hydrochloric acid. 48 cm^3 of hydrogen is collected in the first 30 s. Calculate the mean rate of reaction over this time.
mean rate = volume of product ÷ time = 48 ÷ 30 = **1.6 cm^3/s**.

Worked Example: *(Illustrative data.)* A conical flask containing marble chips and acid loses 0.66 g in 2 minutes. Mean rate = 0.66 ÷ 120 = **0.0055 g/s** (5.5×10^{-3} g/s). If asked for g/min: 0.66 ÷ 2 = 0.33 g/min.

Exam Tip: Always check the time units in the question. If the data are in minutes but the answer is asked "in g/s", convert minutes to seconds first (× 60). Rates are almost always quoted to 2 or 3 significant figures.

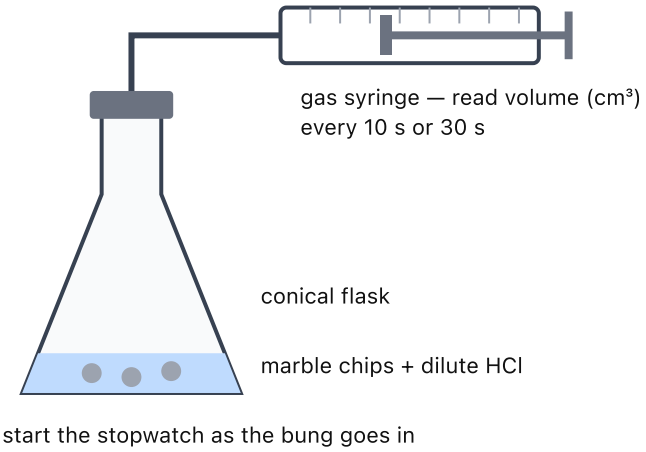
7.2 Practical Methods for Following a Reaction

You need to be able to **suggest a practical method** to determine the rate of a given reaction. The choice depends on what changes as the reaction goes on:

What changes?	Method	Example reaction
A gas is given off	Collect the gas in a **gas syringe** (or an upturned measuring cylinder / burette	$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ $Mg + 2HCl \rightarrow MgCl_2 + H_2$

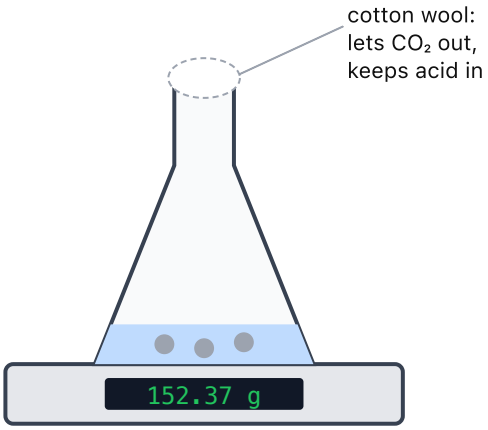
	over water) and record the volume at regular time intervals	
A gas is given off and escapes	Stand the flask on a **top-pan balance**; record the loss in mass at regular intervals (cotton wool plug lets gas out but stops acid spray)	Marble chips + hydrochloric acid (CO_2 is heavy enough to give a measurable mass loss; H_2 is too light)
A precipitate forms, solution goes cloudy	Time how long it takes for a cross drawn under the flask to **disappear** from view (or use a light sensor / colorimeter)	$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$
A colour changes	Time how long a colour takes to appear/disappear; use a **colorimeter** for a quantitative measurement	Iodine clock-type reactions
pH changes	pH probe and data logger	Neutralisation reactions

Method A: gas syringe



start the stopwatch as the bung goes in

Method B: loss in mass



record mass every 30 s; mass falls as CO_2 escapes

7.3 Core Practical — Investigating Rates of Reaction

The specification requires you to investigate the effects of changing the conditions of a reaction on the rate by **(a) measuring the production of a gas** (hydrochloric acid and marble chips) and **(b) observing a colour change / cloudiness** (sodium thiosulfate and hydrochloric acid).

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Part (a): marble chips and hydrochloric acid — effect of surface area

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

1. Measure 50 cm^3 of dilute hydrochloric acid (e.g. 1 mol/dm^3) into a conical flask using a measuring cylinder.
2. Set up a gas syringe (or upturned measuring cylinder full of water in a trough) connected to a bung and delivery tube.
3. Weigh out a fixed mass of **large** marble chips (e.g. 5.0 g), add them to the flask, quickly insert the bung and start the stopwatch.
4. Record the volume of CO_2 every 30 s until the reaction stops (no more gas collected).
5. Repeat with the same mass of **medium** and then **small** chips (or powder), keeping everything else the same.
6. Plot volume of gas (y) against time (x) for each size; compare the initial gradients.

Variables: **independent** — size of marble chips (surface area); **dependent** — volume of CO_2 (and hence rate); **control** — mass of chips, volume and concentration of acid, temperature, type of marble.

To investigate **concentration** instead, use the same mass and size of chips with acid of different concentrations (e.g. 0.5, 1.0, 1.5, 2.0 mol/dm^3).

Part (b): sodium thiosulfate and hydrochloric acid — effect of temperature (or concentration)

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

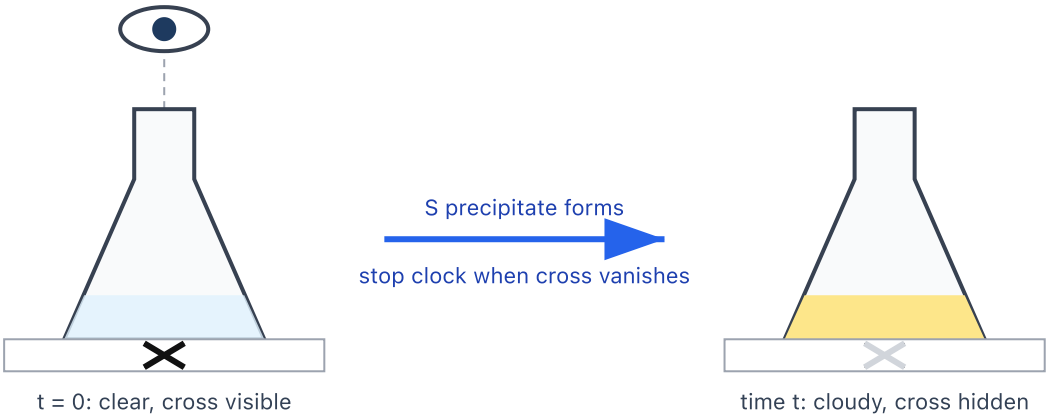
The solid sulfur forms as a pale yellow precipitate, making the solution **cloudy (opaque)**.

1. Draw a black cross on a piece of paper and place a conical flask on top.
2. Add a fixed volume (e.g. 50 cm^3) of sodium thiosulfate solution. Warm it in a water bath to the chosen temperature and record the temperature.
3. Add a fixed volume (e.g. 5 cm^3) of dilute hydrochloric acid, swirl and start the stopwatch.
4. Look down through the solution from above; stop the clock when the cross can **no longer be seen**.

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5. Repeat at different temperatures (e.g. 20, 30, 40, 50 °C). Calculate a relative rate = **1/time**.

The "disappearing cross" method



Temperature / °C	Time for cross to disappear / s	Relative rate = 1/t (s^{-1})
20	120	0.0083
30	62	0.016
40	33	0.030
50	17	0.059

Illustrative data only. The pattern to remember: a rise of about 10 °C roughly halves the time, so roughly doubles the rate.

Exam Tip: Evaluation points for the core practical: the end point of the cross method is **subjective** (different people judge "disappeared" differently) — the same person should judge every run, using the same cross; better still, use a light sensor. In the gas method, some gas escapes before the bung is inserted, and CO_2 is slightly soluble in water if collected over water. The SO_2 produced is toxic — do the thiosulfate experiment in a well-ventilated room and asthmatics should take care.

7.4 Collision Theory

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> For a reaction to happen, reactant particles must **collide** with each other, and the collision must have at least a minimum amount of energy — the **activation energy**. Collisions with enough energy to react are called **successful** (or effective) collisions.

Most collisions are *not* successful: the particles just bounce off each other. The rate of a reaction therefore depends on:

- the **frequency of collisions** (how many collisions there are per second), and
- the **energy of the collisions** (what fraction of collisions have energy ≥ activation energy).

Anything that increases the number of **successful collisions per second** increases the rate of reaction.

> **Exam Tip:** Always say "**more frequent** collisions" or "**more collisions per second**" — not just "more collisions". Over the whole reaction the total number of collisions might be the same; it is the *frequency* that matters for rate.

7.5 Factors Affecting the Rate of Reaction

Temperature

Increasing the temperature increases the rate for **two** reasons:

1. Particles move **faster**, so they collide **more frequently**.
2. Particles have **more energy**, so a **greater proportion** of collisions have energy equal to or greater than the activation energy — more collisions are successful.

The second effect is the more important one, which is why a small temperature rise has such a big effect on rate.

Concentration of a solution

Increasing the concentration means there are **more reactant particles in the same volume**. Particles are closer together, so collisions are **more frequent**. The energy of each collision is not

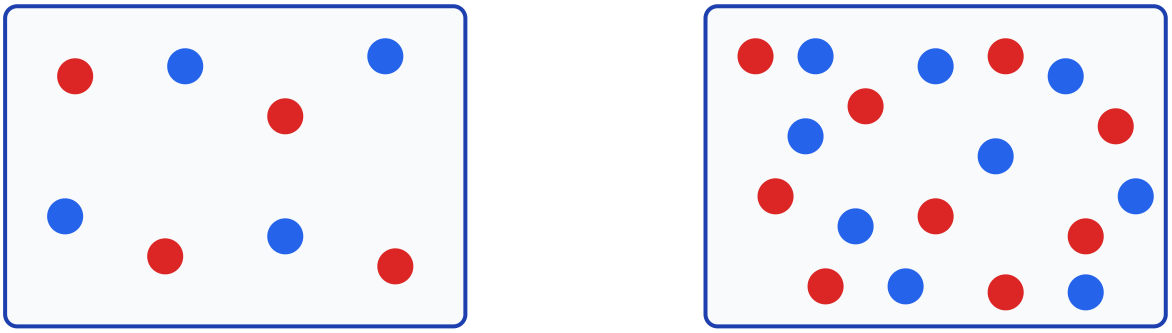
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changed.

Pressure of a gas

Increasing the pressure of a gas (by squeezing the same number of molecules into a smaller volume, or adding more gas) has the same effect as increasing concentration: **more particles per unit volume**, so **more frequent collisions**.

Increasing concentration (or gas pressure)



low concentration: particles far apart

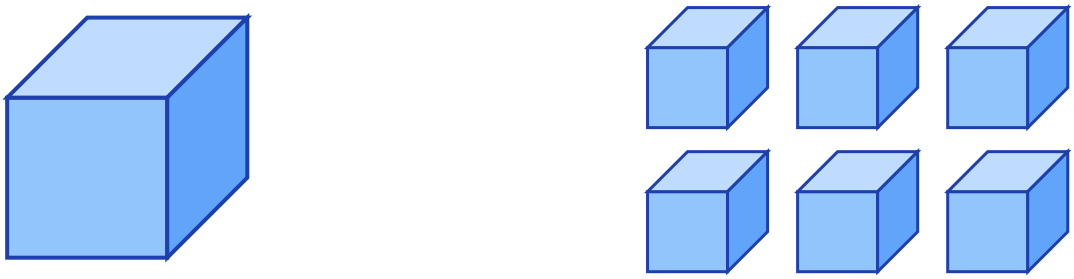
high concentration: more particles per cm³

More particles in the same volume → collisions more frequent → faster rate

Surface area to volume ratio of a solid

Only particles on the **surface** of a solid can collide with particles in the solution or gas. Breaking a solid into smaller pieces (or a powder) **increases the surface area to volume ratio**, exposing more particles, so collisions are **more frequent**.

Cutting a solid increases surface area : volume



2 cm cube: SA = 24 cm², V = 8 cm³

SA : V = 3 : 1

eight 1 cm cubes (6 shown): SA = 48 cm², V = 8 cm³

SA : V = 6 : 1 — twice as much surface exposed

Worked Example: A cube of side 3 cm is cut into 1 cm cubes. Compare the surface area to volume ratios.

Large cube: SA = $6 \times 3^2 = 54$ cm²; V = $3^3 = 27$ cm³; SA : V = 54 : 27 = **2 : 1**.
27 small cubes: SA = $27 \times 6 \times 1^2 = 162$ cm²; V = 27 cm³; SA : V = 162 : 27 = **6 : 1**.
The ratio has tripled, so many more particles are exposed to collisions and the reaction is faster.

Catalysts

A catalyst increases the rate by providing an **alternative reaction pathway with a lower activation energy** (see 7.7). It does not increase the frequency of collisions; instead, a greater proportion of collisions now have enough energy to react.

Summary of the factors

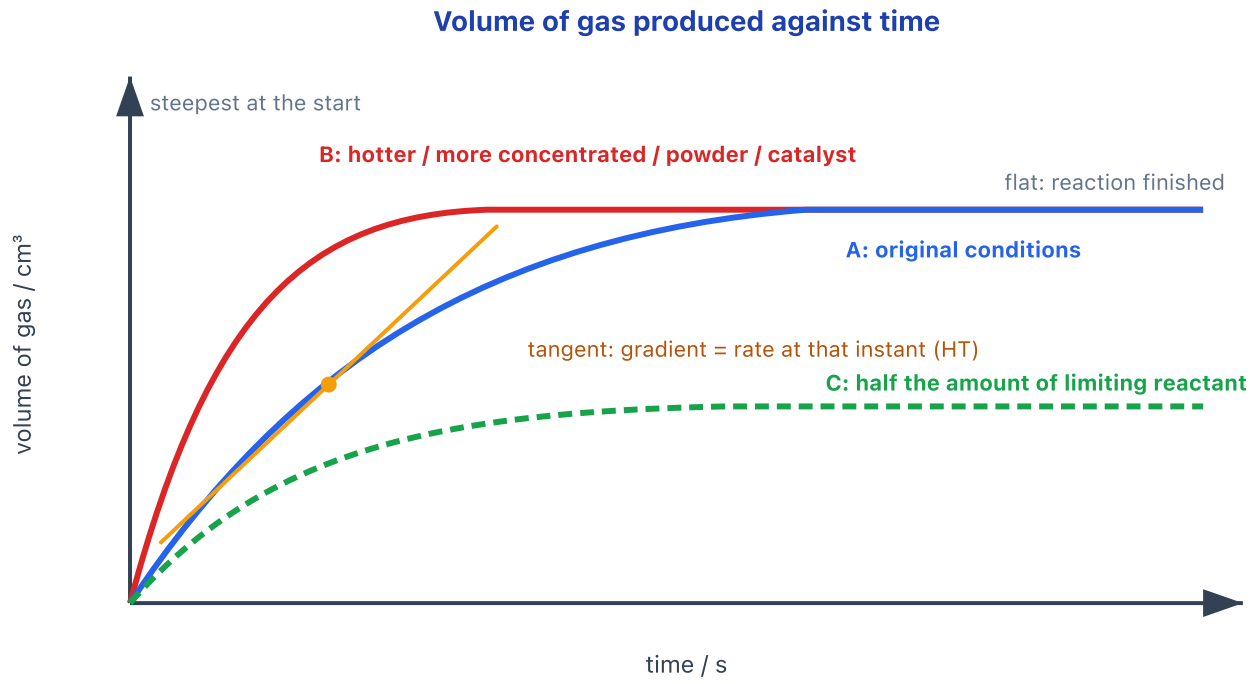
Change	Frequency of collisions	Proportion of collisions with $E \geq E_a$	Effect on rate
Increase temperature	increases	**increases**	increases (strongly)
Increase concentration	increases	no change	increases
Increase gas pressure	increases	no change	increases
Increase surface area : volume (smaller pieces)	increases	no change	increases
Add a catalyst	no change	**increases** (lower E_a)	increases

Exam Tip: A full-mark collision-theory answer usually has a **chain of three steps**: (1) what happens to the particles ("more particles in the same volume" / "particles have more energy"), (2) the effect on collisions ("more frequent collisions" / "more collisions have energy ≥ activation energy"), (3) "so more successful collisions per second, so the rate increases". Don't say particles "move faster" for a concentration change — they don't.

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7.6 Interpreting Rate Graphs

You must be able to **interpret graphs of mass, volume or concentration of reactant or product against time**.



- The **gradient (steepness)** of the curve shows the rate. The steeper the line, the faster the reaction.
- The curve is **steepest at the start**, when the concentration of reactants is highest, so collisions are most frequent.
- The gradient **decreases** as reactants are used up.
- The line becomes **horizontal** when the reaction has stopped because one reactant (the **limiting reactant**) has been used up.
- The **final amount of product** depends only on the amount of limiting reactant, not on the rate. Curve B reaches the same final volume as A, just sooner. Curve C has half the limiting reactant, so only half the final volume.
- For a graph of **mass of flask against time** (mass-loss method) or **concentration of reactant against time**, the curve goes *down* and levels off — the same rules about steepness apply.

Calculating rates from graphs

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Mean rate over a period = change in quantity ÷ time taken (read two points off the curve).

(HT) The rate at a particular instant is found by drawing a **tangent** to the curve at that time and calculating its **gradient**:

rate at time t = gradient of tangent = change in y ÷ change in x

Worked Example: (HT) *(Illustrative data.)* A tangent drawn to a volume–time curve at t = 40 s passes through (10 s, 18 cm^3) and (70 s, 60 cm^3).
gradient = (60 – 18) ÷ (70 – 10) = 42 ÷ 60 = **0.70 cm^3/s**. This is the rate at 40 s.

Exam Tip: (HT) When drawing a tangent, use a ruler and make the line just touch the curve at the chosen point, with the curve bending away equally on both sides. Make the "triangle" as large as possible to reduce reading errors.

Worked Example: *(Illustrative data.)* In a mass-loss experiment the mass of the flask falls from 152.40 g to 151.80 g in the first 60 s, and from 151.80 g to 151.62 g in the next 60 s. Mean rate (0–60 s) = 0.60 ÷ 60 = 0.010 g/s. Mean rate (60–120 s) = 0.18 ÷ 60 = 0.0030 g/s. The rate falls because the acid is being used up, so its concentration decreases and collisions become less frequent.

7.7 Catalysts and Enzymes

A **catalyst** is a substance that **speeds up** the rate of a reaction **without altering the products** of the reaction, and is itself **unchanged chemically and in mass** at the end of the reaction.

- Because it is not used up, only a **small amount** of catalyst is needed, and it can be reused.
- A catalyst does **not** appear in the overall balanced equation (it may be written above the arrow).

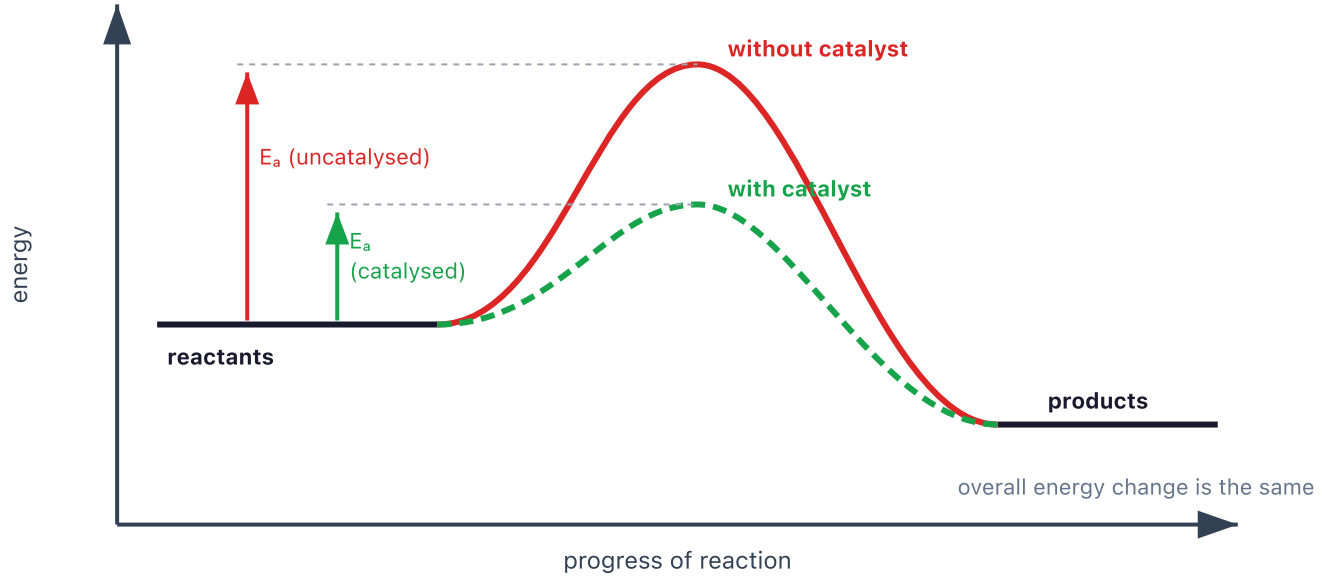
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- A catalyst does **not** change the amount of product made, or the overall energy change of the reaction.
- Different reactions need different catalysts. Many industrial catalysts are **transition metals or their compounds** (e.g. iron in the Haber process; vanadium(V) oxide in the Contact process; platinum/rhodium in catalytic converters).
- Catalysts are economically important: reactions can be run at lower temperatures, saving energy and fuel costs, and making less pollution.

How a catalyst works

A catalyst provides an **alternative reaction pathway** with a **lower activation energy**. At the same temperature, a **greater proportion of collisions** have energy ≥ the (new, lower) activation energy, so there are more successful collisions per second.

Effect of a catalyst on the reaction profile (exothermic reaction)



Enzymes

> **Enzymes** are **biological catalysts** — proteins made by living cells that speed up specific reactions.

- Enzymes in **yeast** catalyse **fermentation**, which turns sugars (e.g. glucose) into ethanol and carbon dioxide. This is used in the **production of alcoholic drinks** such as beer and wine (and

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also in bread-making, where the CO_2 makes the dough rise).

$$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g) \quad \text{(enzymes in yeast)}$$

- Enzymes work best at a particular **optimum temperature** (for yeast fermentation about 30–40 °C). Too cold and the reaction is slow; too hot and the enzyme is **denatured** (its shape changes) and it stops working.
- Enzymes also have an optimum pH.

> **Exam Tip:** Don't say an enzyme is "killed" at high temperature — enzymes are not alive. Say it is **denatured**. Fermentation is covered in more detail (making ethanol solution) in Topic 9.

7.8 Heat Energy Changes in Reactions

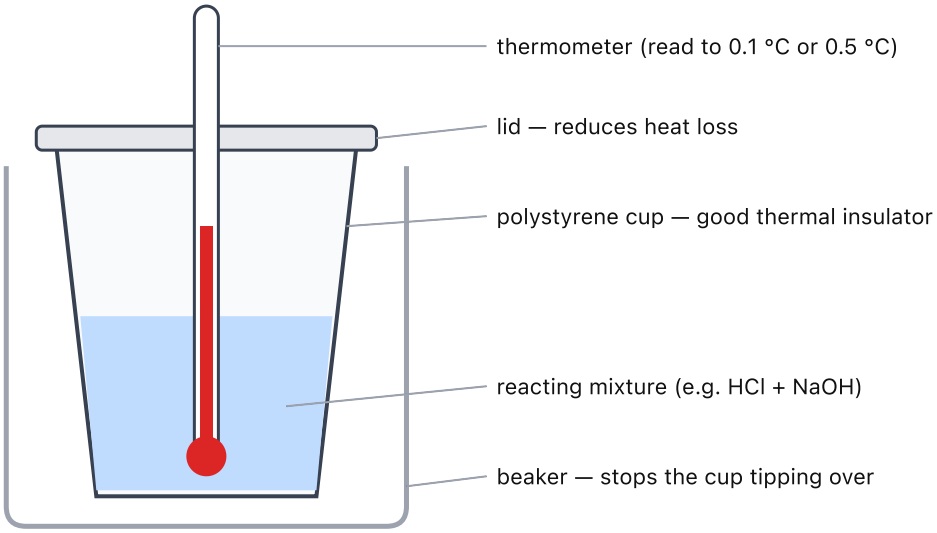
Changes in heat energy accompany many processes. You need to recall that temperature changes happen when:

- **salts dissolve in water** (e.g. ammonium nitrate dissolving — temperature falls; anhydrous copper(II) sulfate or calcium chloride dissolving — temperature rises),
- **neutralisation** reactions take place (acid + alkali — temperature rises),
- **displacement** reactions take place (e.g. zinc + copper(II) sulfate solution — temperature rises),
- **precipitation** reactions take place (two solutions mixed to form an insoluble solid — small temperature change, usually a rise).

These changes can be investigated by **measuring temperature changes** in an insulated **polystyrene cup**.

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Measuring a temperature change with a polystyrene cup



Method (e.g. neutralisation)

1. Measure 25 cm³ of dilute hydrochloric acid into a polystyrene cup standing in a beaker.
2. Record the starting temperature of the acid.
3. Add 25 cm³ of sodium hydroxide solution (at the same starting temperature), put the lid on and stir.
4. Record the **highest** temperature reached (or lowest, for an endothermic change).
5. Temperature change = final temperature – initial temperature. Repeat and calculate a mean.

To compare **displacement** reactions, add a fixed mass of metal powder to a fixed volume of copper(II) sulfate solution; to investigate **dissolving**, add a fixed mass of salt to a fixed volume of water.

> **Exam Tip:** Main source of error: heat is lost to (or gained from) the surroundings, so the measured temperature change is smaller than it should be. Improvements: use a lid, use a polystyrene (not glass) container, stir to mix evenly, read the highest temperature, or take readings over time and extrapolate the cooling curve back to the time of mixing.

Process	Temperature / °C before	Temperature / °C after	Change / °C	Type

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HCl + NaOH	19.5	26.0	+6.5	exothermic
Zn + $CuSO_4$(aq)	20.0	31.5	+11.5	exothermic
NH_4NO_3 dissolving	20.0	13.5	−6.5	endothermic
Citric acid + $NaHCO_3$(aq)	20.5	16.0	−4.5	endothermic

Illustrative data only.

7.9 Exothermic and Endothermic Reactions

> An **exothermic** reaction is one in which **heat energy is given out** to the surroundings — the temperature of the surroundings **increases**.
>
> An **endothermic** reaction is one in which **heat energy is taken in** from the surroundings — the temperature of the surroundings **decreases**.

The "surroundings" include the water the reaction is dissolved in, the container and the thermometer — so a thermometer dipped in an exothermic reaction mixture shows a rise in temperature.

	Exothermic	Endothermic
Heat energy	given out to surroundings	taken in from surroundings
Temperature of surroundings	rises	falls
Energy of products vs reactants	products have **less** energy	products have **more** energy

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Sign of overall energy change (ΔH)	negative (–)	positive (+)
Bond energies	energy released making bonds > energy needed breaking bonds	energy needed breaking bonds > energy released making bonds
Examples	combustion; respiration; neutralisation; most displacement reactions; most precipitation reactions; rusting	thermal decomposition (e.g. $CaCO_3$ → $CaO + CO_2$); photosynthesis; dissolving ammonium nitrate; citric acid + sodium hydrogencarbonate
Uses	self-heating cans, hand warmers	instant cold packs for sports injuries

Exam Tip: In an exothermic reaction the *chemicals* lose energy and the *surroundings* gain it. Be careful with wording: "the reaction gets hot" is fine, but "the reaction takes in heat so it gets hot" is wrong.

7.10 Bond Breaking, Bond Making and Activation Energy

- **Breaking bonds is endothermic** — energy must be supplied to pull atoms apart.
- **Making bonds is exothermic** — energy is released when new bonds form.

In every reaction, bonds in the reactants are broken and new bonds in the products are made. Whether the overall reaction is exothermic or endothermic depends on which is bigger:

- If **more energy is released** making new bonds than is taken in breaking old bonds, the reaction is **exothermic**.
- If **more energy is taken in** breaking bonds than is released making bonds, the reaction is **endothermic**.

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> The **activation energy (E_a)** is the **minimum amount of energy** needed for a reaction to start — the energy colliding particles must have to break bonds and react.

Even exothermic reactions need activation energy: a fuel does not burn until a spark or flame supplies it. Once started, the energy released keeps the reaction going.

7.11 Calculating Energy Changes from Bond Energies (HT)

A **bond energy** is the energy needed to break one mole of a particular covalent bond, measured in **kJ mol^{-1}** (kJ/mol). The same amount of energy is released when one mole of that bond is formed.

energy change = total energy needed to break bonds − total energy released making bonds

$$\Delta H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds made})$$

- A **negative** answer means the reaction is **exothermic**.
- A **positive** answer means the reaction is **endothermic**.

Method

1. Write the balanced equation and **draw out the displayed formulae** so you can see every bond.
2. Count each type of bond in the reactants (remember the coefficients) and add up the energy to break them.
3. Do the same for the products — the energy released.
4. Energy change = (bonds broken) − (bonds made). Include the sign and unit kJ mol^{-1}.

Bond	Bond energy / kJ mol^{-1}	Bond	Bond energy / kJ mol^{-1}
H–H	436	C–H	412
Cl–Cl	243	O=O	498

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H–Cl	432	C=O	805
N≡N	945	O–H	464
N–H	391	C–C	348

Typical data-book values; exam questions always give you the values to use.

Worked Example: (HT) Calculate the energy change for: $H_2 + Cl_2 \rightarrow 2HCl$
Bonds broken: 1 × H–H + 1 × Cl–Cl = 436 + 243 = **679 kJ**
Bonds made: 2 × H–Cl = 2 × 432 = **864 kJ**
Energy change = 679 − 864 = **−185 kJ mol^{-1}** → negative, so **exothermic**.

Worked Example: (HT) Calculate the energy change for the complete combustion of methane: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Displayed: H–C(–H)(–H)–H + 2 O=O → O=C=O + 2 H–O–H
Bonds broken: 4 × C–H = 4 × 412 = 1648; 2 × O=O = 2 × 498 = 996; total = **2644 kJ**
Bonds made: 2 × C=O = 2 × 805 = 1610; 4 × O–H = 4 × 464 = 1856; total = **3466 kJ**
Energy change = 2644 − 3466 = **−822 kJ mol^{-1}** (exothermic — methane is a good fuel).

Worked Example: (HT) Calculate the energy change for the Haber process: $N_2 + 3H_2 \rightarrow 2NH_3$
Bonds broken: 1 × N≡N + 3 × H–H = 945 + 3(436) = 945 + 1308 = **2253 kJ**
Bonds made: 2 NH_3 contain 6 N–H bonds = 6 × 391 = **2346 kJ**
Energy change = 2253 − 2346 = **−93 kJ mol^{-1}** (exothermic).

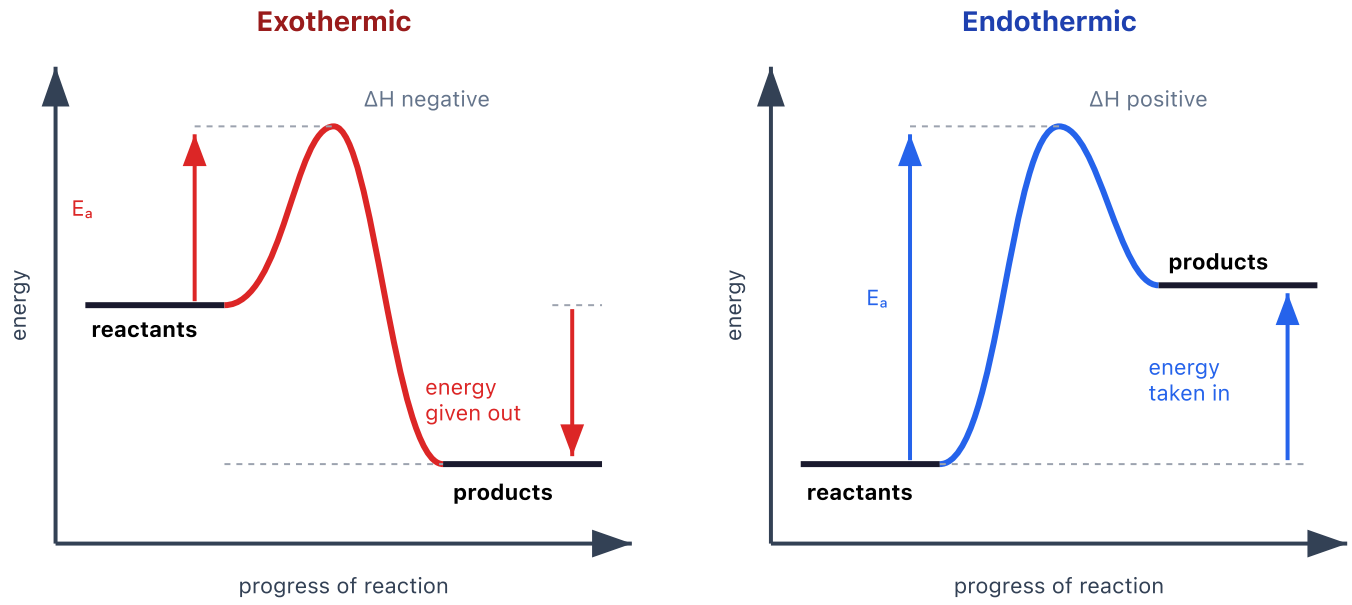
Exam Tip: The most common errors are (1) forgetting to multiply by the coefficients in the equation, (2) miscounting bonds — e.g. CO_2 has **two** C=O bonds and each H_2O has **two** O–H bonds, and (3) doing "made − broken", which gives the wrong sign. Always write "broken − made".

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7.12 Reaction Profiles

A **reaction profile** (energy level diagram) shows the energy of the reactants and products and how energy changes as the reaction proceeds. You must be able to **draw and label** reaction profiles for exothermic and endothermic reactions, showing:

- the **reactants** and **products** (as horizontal lines, labelled with names or formulae),
- the **activation energy** — an arrow from the reactants up to the peak of the curve,
- the **overall energy change** — an arrow from the reactants level to the products level.



Exam Tip: The activation energy arrow always starts at the **reactants** line and ends at the **top of the hump** — not from the products, and not from the x-axis. The energy-change arrow points **down** for exothermic and **up** for endothermic. Label the axes "energy" and "progress of reaction".

Linking it all together

On the reaction profile, going up the hump represents the energy needed to **break bonds** in the reactants (endothermic); coming down the other side represents energy released as **new bonds form** (exothermic). The difference in height between reactants and products is the overall energy change you calculated from bond energies. A catalyst lowers the hump but does not move the reactant or product lines.

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Worked Example: The energy change for $H_2 + Cl_2 \rightarrow 2HCl$ is −185 kJ mol^{-1} and the activation energy is (illustratively) +250 kJ mol^{-1}. Sketch the profile.
Draw reactants ($H_2 + Cl_2$) higher than products (2HCl) by 185 units; draw the hump 250 units above the reactants; label E_a (reactants → peak, upward arrow) and the energy change (reactants → products, downward arrow, −185 kJ mol^{-1}).

7.13 Rates in Everyday Life and Industry

Situation	Factor used	Explanation
Food kept in a fridge lasts longer	lower temperature	slows reactions (and microbial enzymes) that cause decay
Powdered fuels / flour dust can explode	large surface area	very fast combustion — a hazard in flour mills and coal mines
Indigestion tablets are chewed or soluble	surface area	neutralise stomach acid faster
Catalytic converters in cars	catalyst (Pt, Rh, Pd)	speed up conversion of CO and NO_x to CO_2 and N_2 at exhaust temperatures
Haber process	catalyst (iron), compromise temperature, high pressure	acceptable rate without wasting energy (see Topics 4 and 5)
Brewing	enzymes in yeast at optimum temperature	fermentation of sugars to ethanol

Key Points Summary

- Rate = amount of reactant used (or product formed) ÷ time; units cm^3/s, g/s or $mol/dm^3/s$; for "time to a fixed point" experiments, relative rate = 1/t.

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- Choose a method to suit the reaction: gas syringe, mass loss, disappearing cross / colorimeter, pH probe.
- **Core practical:** marble chips + HCl (gas volume, surface area or concentration) and sodium thiosulfate + HCl (cloudiness, temperature or concentration). Know the variables and sources of error.
- Collision theory: particles must collide with energy ≥ activation energy; rate depends on frequency and energy of collisions.
- Higher concentration, gas pressure or surface area : volume → more frequent collisions. Higher temperature → more frequent *and* more energetic collisions.
- On a product–time graph: gradient = rate; steepest at start; flat when the limiting reactant is used up; final amount depends only on the amount of limiting reactant.
- **(HT)** Rate at an instant = gradient of the tangent to the curve.
- A catalyst speeds up a reaction without altering the products and is unchanged in mass and chemically at the end; it provides an alternative pathway with lower activation energy.
- Enzymes are biological catalysts; yeast enzymes ferment sugars to make alcoholic drinks.
- Dissolving salts, neutralisation, displacement and precipitation all involve heat changes — measure with a polystyrene cup, lid and thermometer.
- Exothermic: heat given out, surroundings get hotter, ΔH negative. Endothermic: heat taken in, surroundings get colder, ΔH positive.
- Bond breaking is endothermic; bond making is exothermic.
- **(HT)** Energy change from bond energies = bonds broken − bonds made (kJ mol^{-1}).
- Activation energy is the minimum energy needed to start a reaction.
- Be able to draw and label reaction profiles: reactants, products, E_a (reactants → peak) and the overall energy change.

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